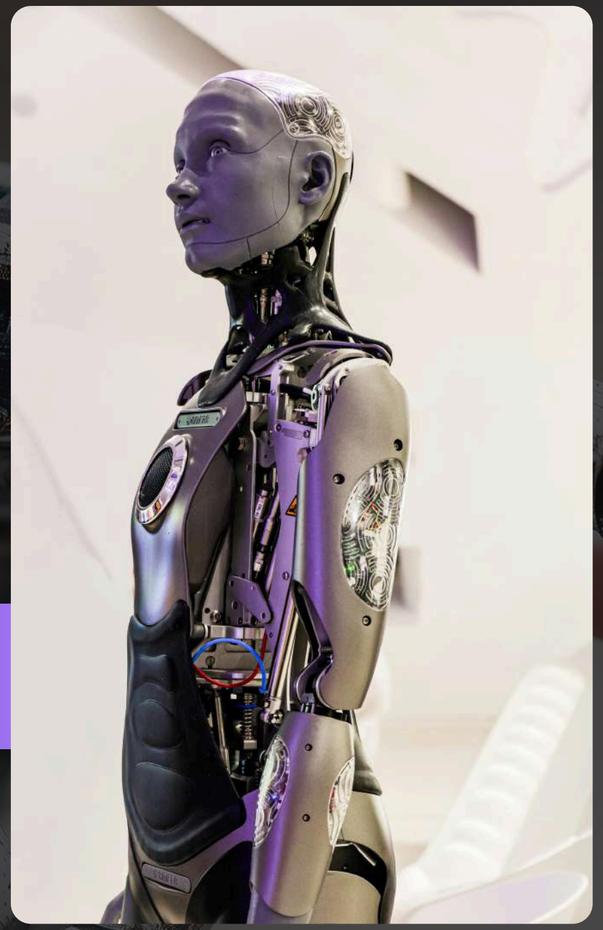
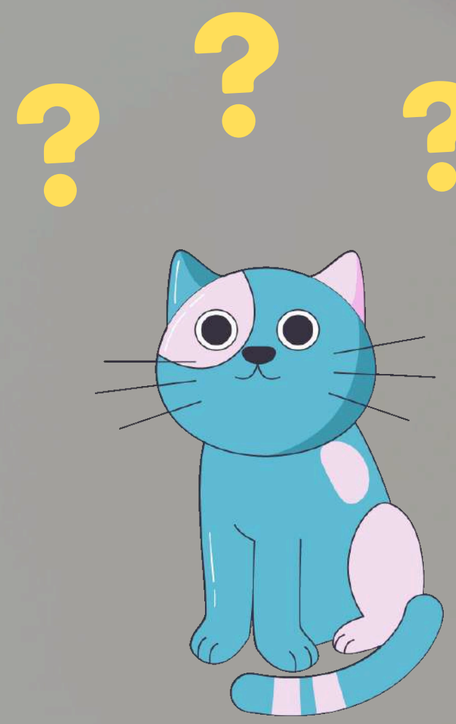


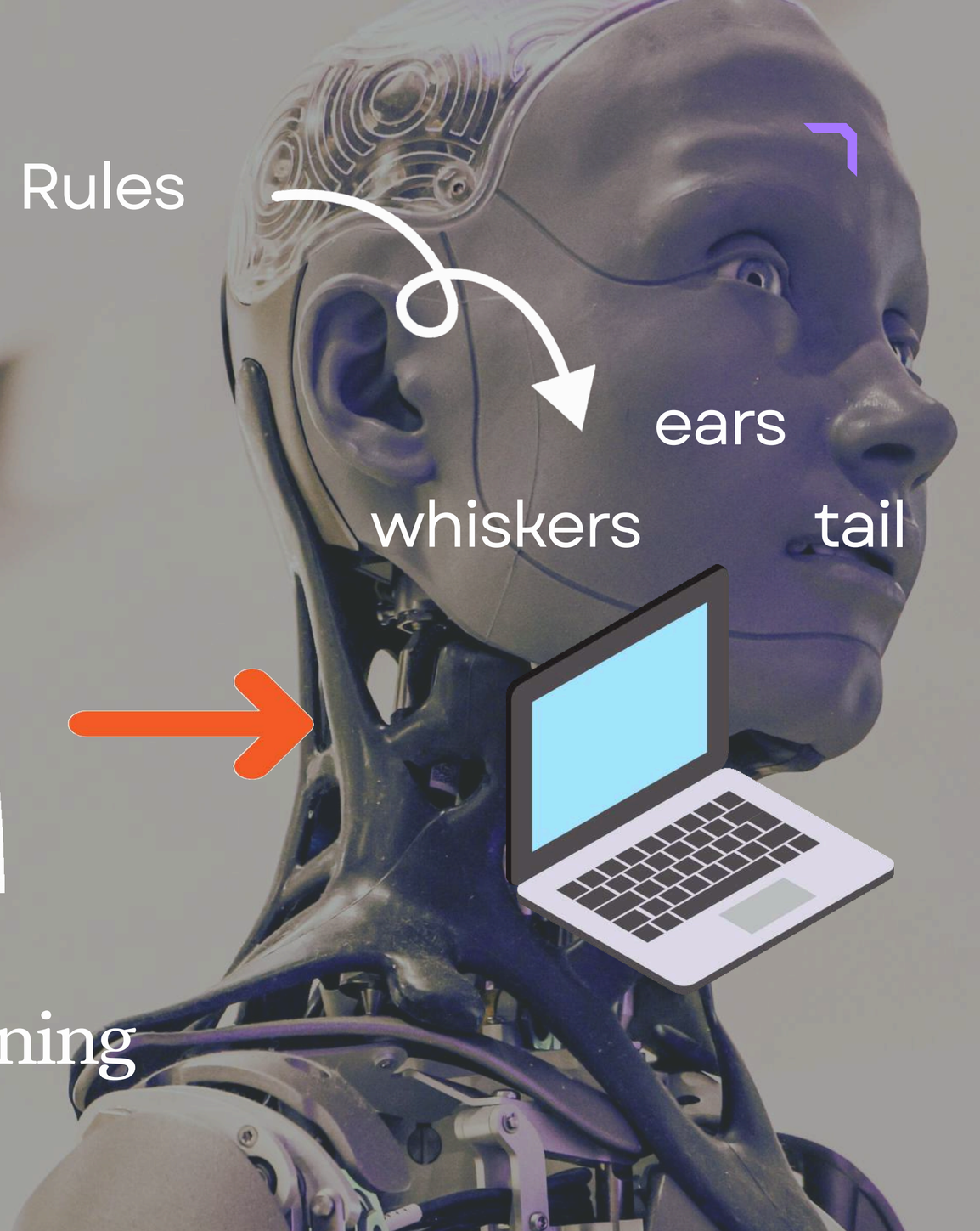
UNIT 1

Deep Learning

Foundations of Deep Learning

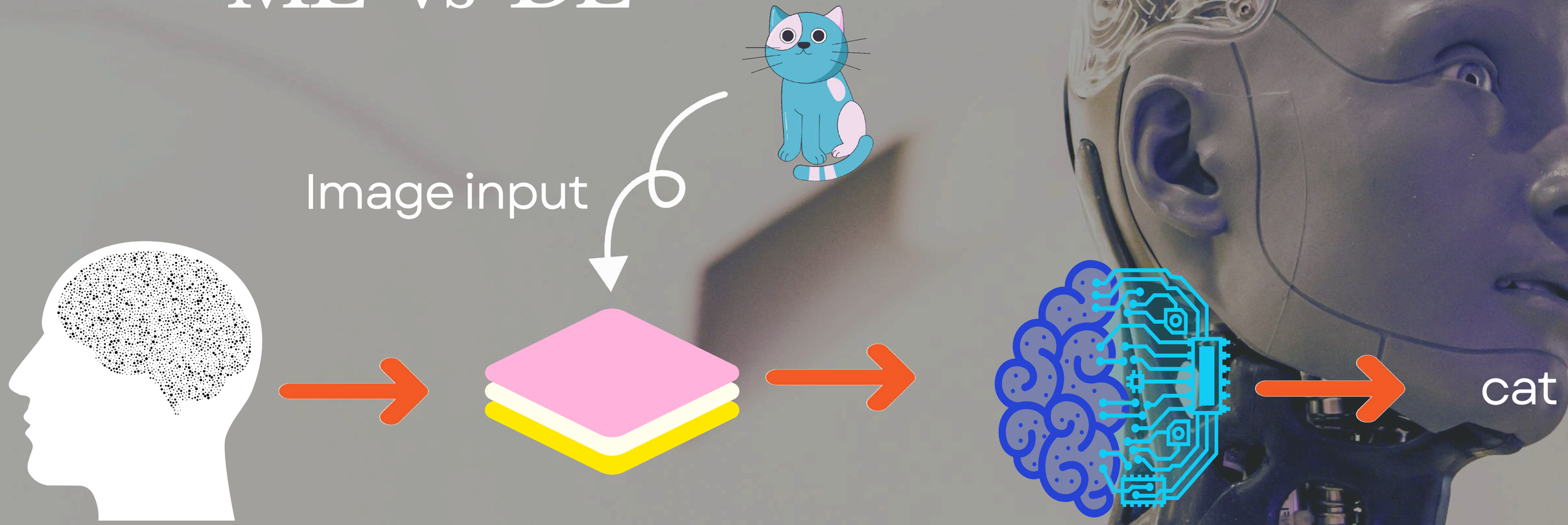
ML Vs DL





Machine Learning

ML Vs DL



Deep Learning

ML Vs DL

Machine Learning (ML)	Deep Learning (DL)
Subset of AI that learns from data using algorithms	Subset of ML that uses neural networks with many layers
Works well with small to medium datasets	Requires very large datasets
Needs manual feature engineering	Automatically extracts features
Faster to train	Takes longer to train
Less computational power required	Requires high computational power (GPU/TPU)
Algorithms: Decision Trees, SVM, Linear Regression	Algorithms: CNN, RNN, Transformers
Easier to interpret results	Harder to interpret (black box)
Suitable for structured data	Best for unstructured data (images, audio, text)

Limitations of ml

1. Requires Large Amounts of Data

- ML models perform well only when trained on vast amounts of high-quality data.
- If the dataset is small, the model may not generalize well.

2. Computationally Expensive

- Training ML models, especially deep learning models, requires powerful hardware such as GPUs and TPUs.
- This increases cost and resource usage.

Limitations of ml

3. Overfitting and Underfitting

- **Overfitting:** Model learns too much from training data (including noise) and performs poorly on new data.
- **Underfitting:** Model is too simple and fails to capture important patterns.

4. Requires Domain Expertise

- Choosing the right algorithm, preprocessing data, and tuning hyperparameters require expert knowledge.

Correlation between AI, ML & Deep Learning

- Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) are closely related concepts.
- Each field is a subset of the previous one:
  - AI → broad field
  - ML → subset of AI
  - DL → subset of ML

Artificial Intelligence (AI)

Correlation between AI, ML & Deep Learning

- Broad field that enables machines to mimic human intelligence
- Includes reasoning, problem-solving, decision-making, and learning

Correlation between AI, ML & Deep Learning

Machine Learning (ML)

- Subset of AI that allows machines to learn from data
- Does not require explicit programming for every task
- Improves performance with experience

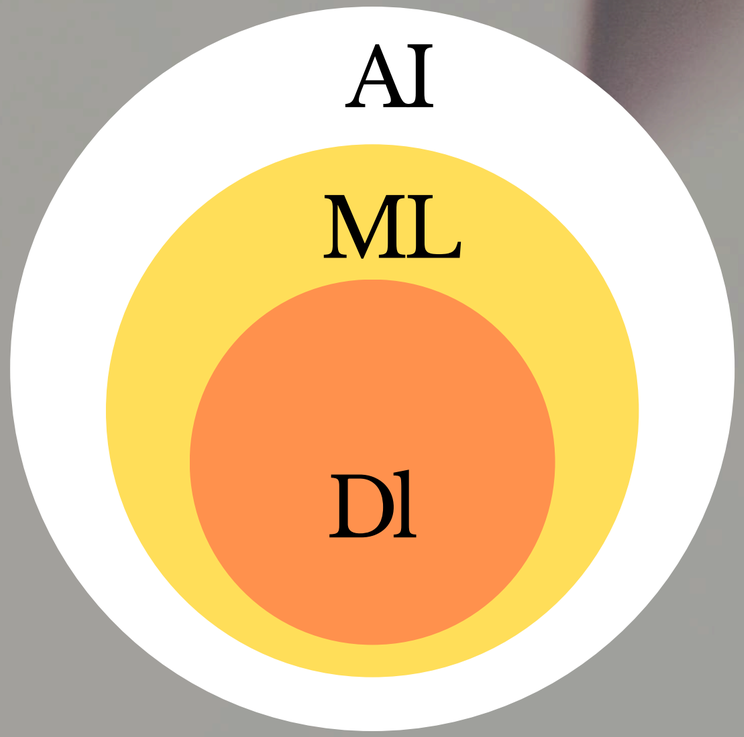
Correlation between AI, ML & Deep Learning

Deep Learning (DL)

- Specialized subset of ML
- Uses neural networks with multiple layers
- Designed to learn from large and complex datasets

Correlation between AI, ML & Deep Learning

Diagram



Deep Learning

Definition

- Deep Learning is a specialized branch of Machine Learning
- Uses artificial neural networks with multiple layers
- Designed to analyze and learn patterns from large datasets

Deep Learning

How It Works

- Mimics how the human brain processes information
- Automatically extracts important features from data
- Improves performance as more data is provided

Deep Learning

Advantages

1. High Accuracy
   - Performs very well in tasks like image recognition and speech processing

2. Automatic Feature Extraction
   - No need for manual feature selection unlike traditional ML

Deep Learning

Disadvantages

1. Requires Large Data
   - Works best with massive datasets
   - Small datasets may give poor results

2. High Computational Cost
   - Needs powerful hardware like GPUs/TPUs

Industrial Tools for Deep Learning

TensorFlow

- Developed by Google
- One of the most powerful and widely used deep learning frameworks
- Supports training neural networks on CPUs and GPUs
- Highly scalable and flexible
- Used in industries like healthcare, finance, and automation

Industrial Tools for Deep Learning

PyTorch

- Developed by Meta (Facebook)
- Popular deep learning library known for flexibility and ease of use
- Uses dynamic computational graphs
- Makes debugging and model development intuitive
- Widely used in research and academic projects

Industrial Tools for Deep Learning

Keras

- High-level deep learning API
- Runs on top of TensorFlow
- Designed to be simple and beginner-friendly
- Ideal for rapid prototyping
- Allows developers to build models quickly without complex coding

Industrial Tools for Deep Learning

OpenCV

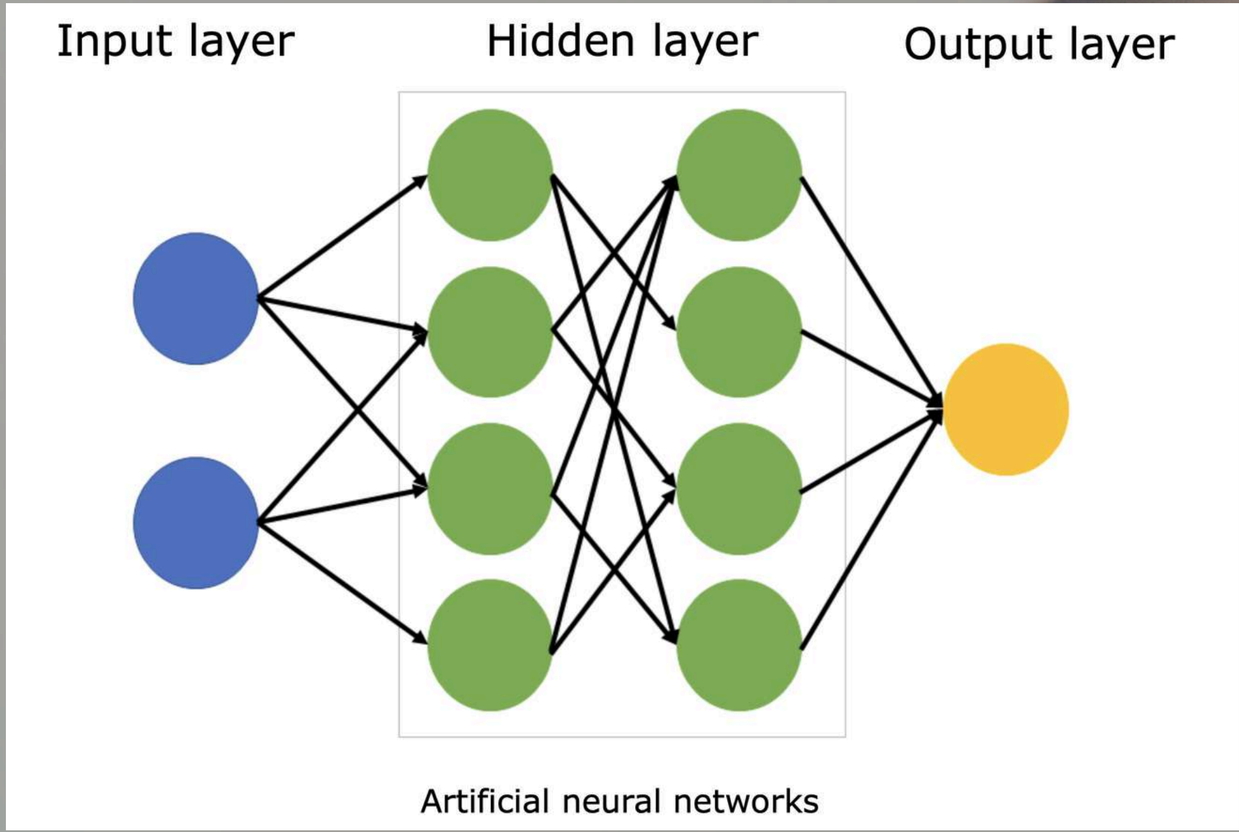
- Open-source computer vision library
- Mainly used for image and video processing tasks
- Supports deep learning integration

Common in real-time applications like:

- Face recognition
- Object detection

Working of Deep learning

Diagram of NN



Working of Deep learning

1. Input Layer

Receives raw input data

- Data can be:
- Images
- Text
- Audio
- Numerical values

Deep learning generally requires large amounts of data

Working of Deep learning

2. Hidden Layers

- Consist of multiple layers of neurons
- Each layer detects specific features
  - First layer → simple patterns (edges, colors)
  - Middle layers → shapes or textures
  - Final hidden layers → complex patterns
- Features are gradually combined to understand the data

Working of Deep learning

3. Output Layer

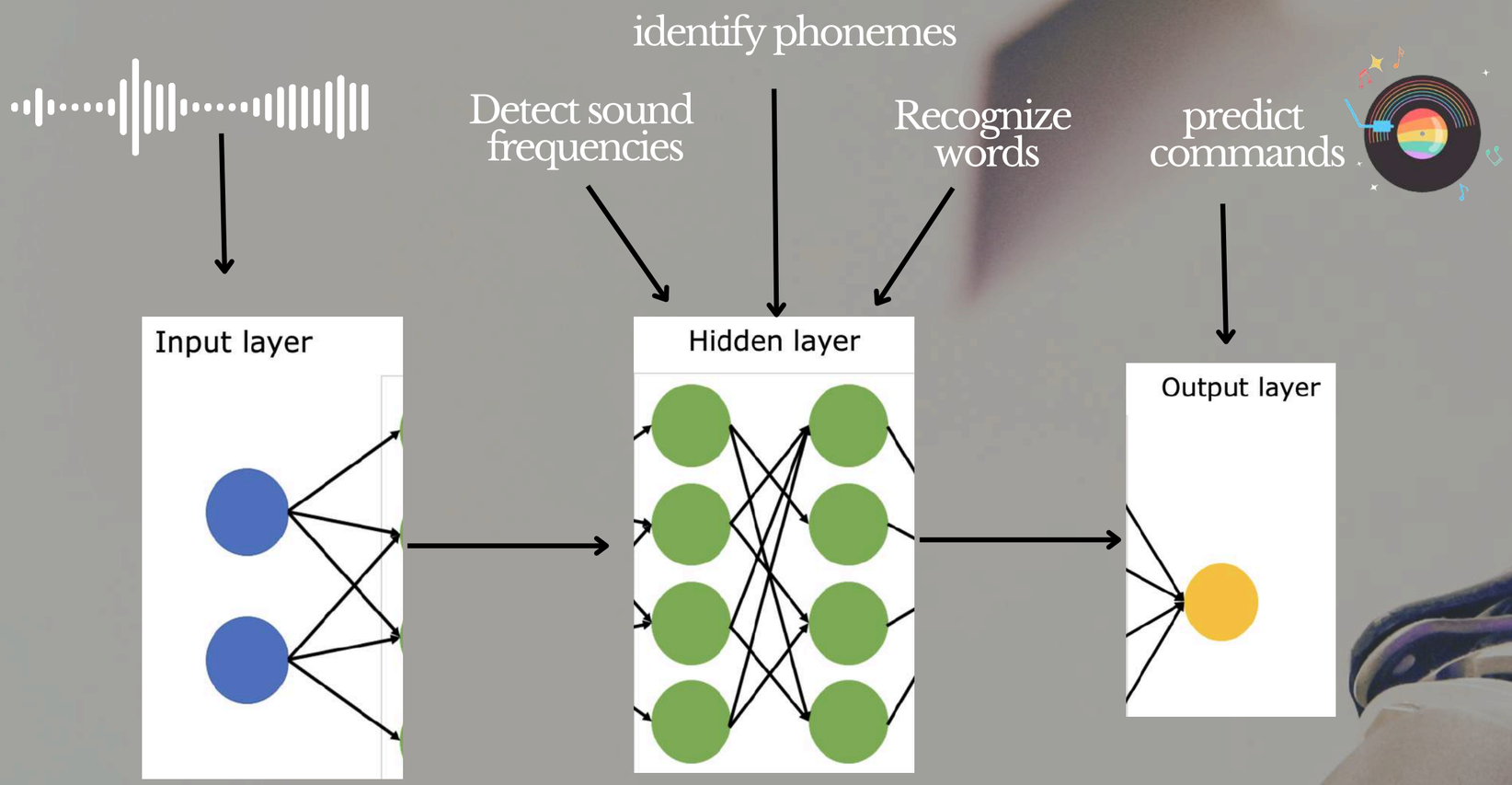
- **Produces the final prediction or result**

Example outputs:

- **Classifying an object**
- **Predicting a value**
- **Recognizing speech**

Working of Deep learning

Example



Common Architectural Principles of Deep Learning

- Deep neural networks follow specific design principles
- These principles ensure efficient learning and accurate predictions

Common Architectural Principles of Deep Learning

1. Layered Structure

Deep networks consist of multiple layers:

- Input layer
- Several hidden layers
- Output layer

Each layer processes data and passes it forward

Common Architectural Principles of Deep Learning

2. Activation Functions

- Introduce non-linearity into the model
- Help networks learn complex patterns
- Common activation functions:
  - ReLU
  - Sigmoid
  - Tanh

Common Architectural Principles of Deep Learning

3. Weights and Biases

- Every neuron has:
- Weights → determine importance of input features
- Bias → adjusts output for better learning
- These parameters are updated during training

Common Architectural Principles of Deep Learning

4. Loss Function & Optimization

- Loss function measures difference between predicted and actual values
- Goal → Minimize loss
- Optimization algorithms adjust weights:
- Gradient Descent
- Adam

Common Architectural Principles of Deep Learning

5. Regularization & Dropout

- Prevents overfitting
- Techniques include:
- Dropout → randomly disables neurons during training
- L2 Regularization → penalizes large weights

underfitting , overfitting & regularization

UnderFitting

- Model is too simple to learn data patterns
- Performs poorly on training and testing data
- High bias, low variance

underfitting , overfitting & regularization

OverFitting

- Model learns training data too well, including noise
- Performs well on training data but poorly on new data
- Low bias, high variance

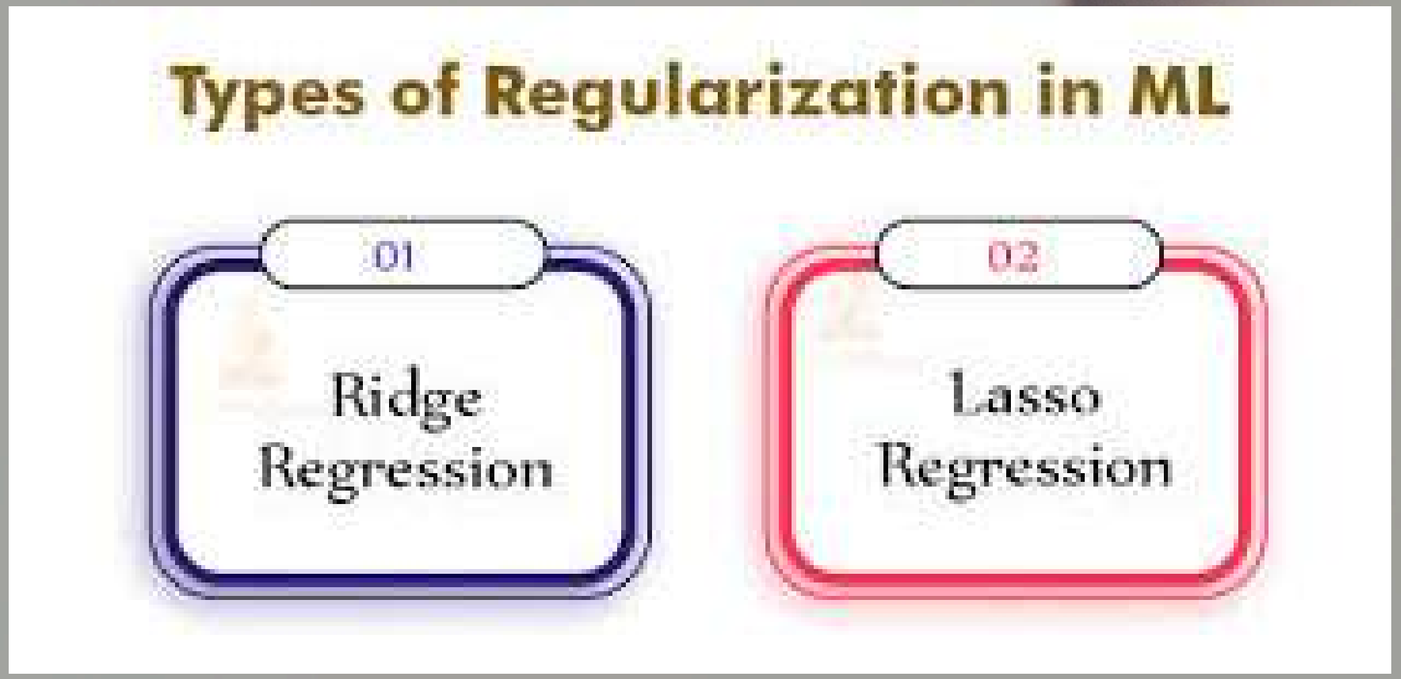
underfitting , overfitting & regularization

Regularization

- Technique used to prevent overfitting
- Adds a penalty term to loss function
- Encourages model to learn general patterns, not noise

underfitting , overfitting & regularization

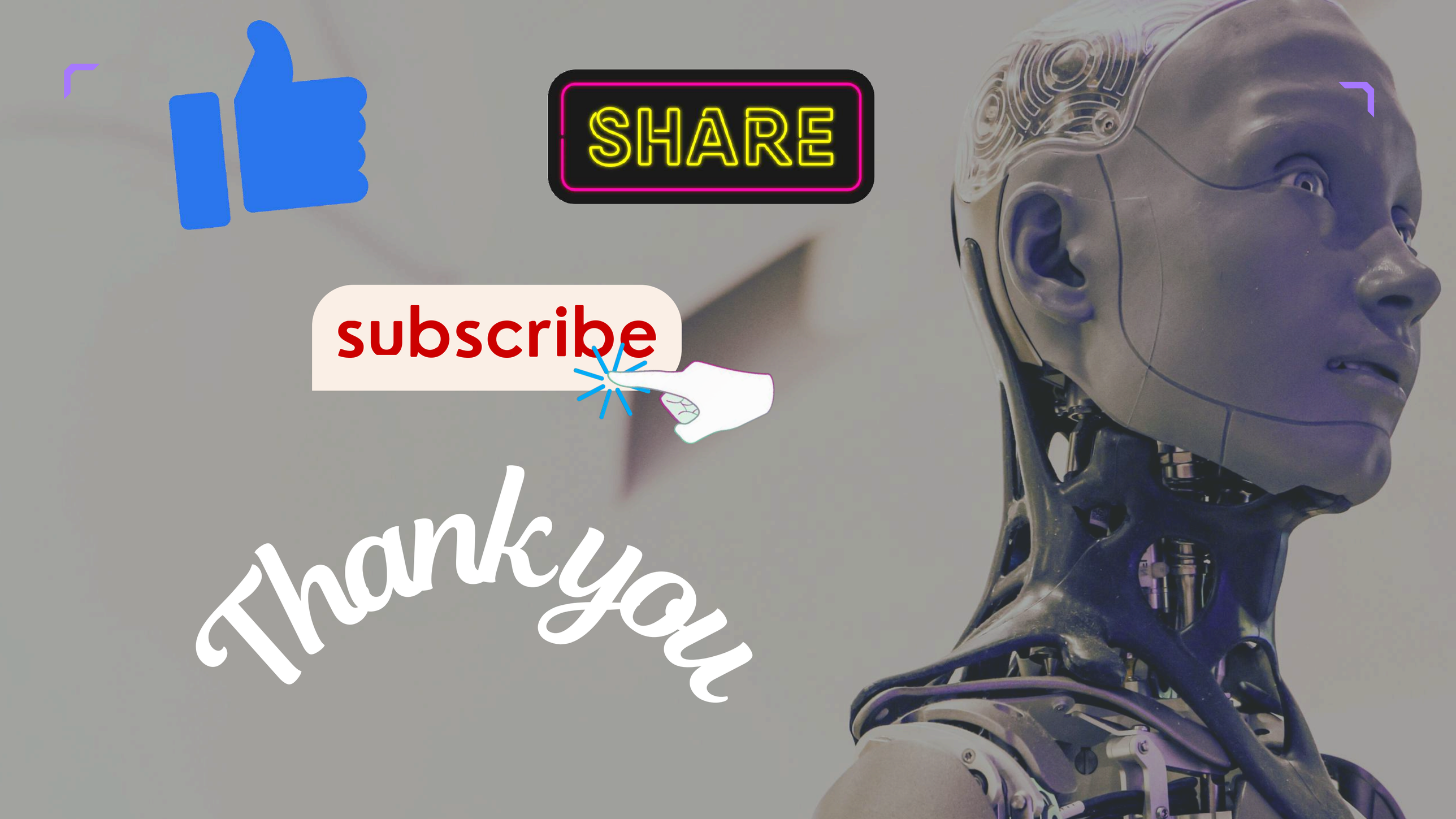
Regularization



Supervised and unsupervised learning

Supervised Learning	Unsupervised Learning
Uses **labeled data**	Uses **unlabeled data**
Learns from **input → output pairs**	Finds **hidden patterns** in data
Goal is **prediction**	Goal is **pattern discovery**
Requires **training dataset with correct answers**	Does **not need correct answers**
Examples: Classification, Regression	Examples: Clustering, Association
More accurate when labeled data is available	Useful when labeling is difficult or costly
Teacher guides learning	No teacher — model learns itself

Real life Applications of DL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